localCloudManagement

Service Description

Service ID: *"localCloudManagement"*

Abstract
This document describes the `localCloudManagement` service, version 1.0.0, produced by the LocalCloudManager support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `localCloudManagement` service, which provides the operator console API of the local cloud: overview, registry and data views, orchestration rules, consumers, inter-cloud state, sysMonitor history and subscriptions, plant descriptions, logs and container lifecycle. It is produced by the LocalCloudManager system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

New in this workspace.

1.2 How This Service Is Meant to Be Used

The browser console, served by LocalCloudManager on its console connector (8490, host loopback), calls the operations after the operator has logged in with the console password. Arrowhead systems call the same operations on the Arrowhead connector (8491) with their certificate and identity token. Each view is assembled on request from the providing systems with LocalCloudManagerś own identity.

1.3 Important Delimitations and Dependencies

Registered with `generic_https` and policy `CERT_AUTH` on the Arrowhead connector: there a caller needs its client certificate and an identity token of the same system, and the operation is allowed for Sysop, the systems of `lcm.api.whitelist` and systems with a ConsumerAuthorization grant of the operation. On the console connector the console session authenticates instead; `session`, `login` and `logout` exist only there and are not registered. Changing requests need `X-Arrowhead-Action: confirm` (428 otherwise). The registered operation names are camelCase, which deviates from the kebab-case convention of ARROWHEAD-NAMING.md.

2 Service Interface

This section describes the interfaces to the `localCloudManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.



Figure 1: SysML-style block of the `localCloudManagement` service and its operations.

The following interface operations are available.

2.1 operation session () : Session

Returns whether the console session is valid (no session needed; console connector only, not registered).

2.2 operation login (Login)

Logs in with the console password; sets the HttpOnly, SameSite=Strict session cookie (console connector only, not registered).

2.3 operation logout ()

Ends the console session (console connector only, not registered).

2.4 operation overview () : View

Overview: containers, registered systems, brokers, sysMonitor flows, activity.

2.5 operation serviceRegistryManagement () : View

Registry inventory: devices, systems, services and interface templates.

2.6 operation dataManager () : View

DataManager historian streams and recent records.

2.7 operation consumers () : View

Per provider service, the authorized and the orchestrated consumers.

2.8 operation interCloud () : View

Relay connection, Gatekeeper and GateTunnel sessions.

2.9 operation sysMonitorHistory () : View

Monitored systems and their history.

2.10 operation sysMonitorSubscriptions () : View

SysMonitor providers, subscriptions and intake state.

2.11 operation subscribeSysMonitor (SysMonitorSubscriptionRequest) : View

Sets the SysMonitor subscription of a provider (appended to the path).

2.12 operation unsubscribeSysMonitor (Name)

Removes the SysMonitor subscription of a provider.

2.13 operation plantDescriptions () : View

Plant descriptions with the registered systems and services offered as choices.

2.14 operation createPlantDescription (PlantDescription) : PlantDescription

Creates a plant description in the PlantDescriptionEngine.

2.15 operation replacePlantDescription (PlantDescription) : PlantDescription

Replaces a plant description (identifier appended).

2.16 operation updatePlantDescription (PlantDescription) : PlantDescription

Updates a plant description (identifier appended).

2.17 operation removePlantDescription (Identifier)

Removes a plant description (identifier appended).

2.18 operation orchestrationRules () : View

SimpleStore rules and the dynamic orchestrations of DynamicServiceOrchestration.

2.19 operation createOrchestrationRule (OrchestrationRule) : View

Creates a SimpleStore rule.

2.20 operation removeOrchestrationRule (Identifier)

Removes a SimpleStore rule (identifier appended).

2.21 operation logs () : LogLines

Log lines of a container of the cloud.

2.22 operation lifecycle (String) : LifecycleResult

Starts, stops or restarts a managed container: `POST /api/containers/<service>/<action>`.

Console operations outside the registered operation list

The console API also offers the following operations. They are not part of the `operations` property registered in the ServiceRegistry (the console is their consumer); they are served on both connectors with the same access rules as the registered operations (on the Arrowhead connector only for Sysop and `lcm.api.whitelist`, since no grant can name them) and, for changes, need the confirmation header.

2.23 operation features () : Features

Which optional console views are enabled.

2.24 operation authorizationPolicies () : PolicyView

Local-cloud management policies of ConsumerAuthorization (editable) and provider-owned local policies (read-only).

2.25 operation grantAuthorizationPolicy (PolicyChange) : ManagementPolicy

Grants a local management policy, or replaces the policy of the same provider and target when `expected` matches the stored one. Provider and service must be registered in the local cloud; policy types are `ALL`, `WHITELIST` and `BLACKLIST`.

2.26 operation revokeAuthorizationPolicy (PolicyChange) : Removal

Revokes a local management policy (instance identifier appended to the path) when `expected` matches the stored policy.

2.27 operation changeOrchestrationRulePriority (RulePriorityChange) : View

Changes the priority of a SimpleStore rule (identifier appended) with SimpleStore `modify-priorities`, or by create and remove when `ORCHESTRATION_PRIORITY_MODE=replace`.

2.28 operation sosConfigurations () : View

SoS compositions (plant descriptions of the PlantDescriptionEngine); one composition when its identifier is appended.

2.29 operation createSoS (SoSChange) : PlantDescription

Stores a validated SoS composition (unique systems and ports, required-to-provided connections, included compositions without cycles).

2.30 operation replaceSoS (SoSChange) : PlantDescription

Replaces an SoS composition (identifier appended) when `expected` matches the stored one.

2.31 operation removeSoS (SoSChange) : Removal

Removes an SoS composition (identifier appended) when `expected` matches the stored one.

2.32 operation localCloudConfiguration (String) : View

Local-cloud configuration view (404 when `lcm.local-cloud-configuration.enabled=false`): the route below `/api/localcloud` selects one operation of the `localCloudConfiguration` service of LocalCloudConfigurator [3], which is consumed through orchestration with LocalCloudManagerś certificate and identity token: `GET` of `/catalog`, `/status`, `/configurations`, `/configurations/{uuid}` and `/deployments/{id}`; `POST` of `/plan`, `/deployments` and `/configurations`; `PUT` and `DELETE` of `/configurations/{uuid}`. Status and body of the configurator are returned unchanged; other routes are 404.

2.33 operation chipletFloorplans (String) : View

Chiplet floorplan view when `lcm.chiplet-floorplans.enabled=true` (404 otherwise), through orchestrated ChipletFloorplanGenerator services: `GET` list (`page`, `size` up to 200), `/library` and `/{id}`; `POST` create (stored inactive, 201) and `POST /check-network`; `PUT /{id}` with `expectedUpdatedAt` (409 when changed); `DELETE /{id}`. Bodies carry `floorplan` with 1–64 chiplets and at most 1024 links.

3 Information Model

Here, all data objects that can be part of the `localCloudManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.16, which are used to represent things like hashes and identifiers.

3.1 struct LifecycleResult

Field	Type	Description
service	String	Compose service.
action	String	`start`, `stop` or `restart`.
accepted	Boolean	Forwarded to the Docker proxy.
at	String	Time of the action.

3.2 struct LogLines

Field	Type	Description
service	String	Compose service.
lines	List<String>	Log lines (`tail` latest).
collectedAt	DateTime	Collection time.

3.3 struct Login

Field	Type	Description
password	String	Console password (JSON only; 204 on success, 401 wrong, 429 locked).

3.4 struct OrchestrationRule

Field	Type	Description
consumer	Name	Consumer system.
serviceInstanceId	Identifier	Service instance.
priority	Number	Priority (1 highest).

3.5 struct PlantDescription

Field	Type	Description
*	Object	Plant description as in PlantDescriptionEngine `plantDescriptionManagement`.

3.6 struct Session

Field	Type	Description
authenticated	Boolean	Whether the request carries a valid console session.
configured	Boolean	Whether a console password is configured.

3.7 struct SysMonitorSubscriptionRequest

Field	Type	Description
observationTypes	List<String>	Types to receive, `*` for all.
enabled	Boolean	Default true.

3.8 struct View

Field	Type	Description
*	Object	View document assembled from the providing systems; a failing source is reported in the document instead of failing the request.

3.9 struct Features

Field	Type	Description
localCloudConfiguration	Boolean	Local-cloud configuration view enabled.
chipletFloorplans	Boolean	Chiplet floorplan view enabled.
managerName	Name	LocalCloudManager, or ChipletLocalCloudManager with floorplans.

3.10 struct ManagementPolicy

Field	Type	Description
instanceId	Identifier	Policy instance (assigned by Consumer Authorization).
provider	Name	Registered local provider.
targetType	String	`SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Service definition or event type.
description	String	At most 1000 characters.
defaultPolicy	Object	`policyType` (`ALL`, `WHITELIST`, `BLACKLIST`) and `policyList` of local system names.
scopedPolicies	Object	Per operation, a policy like `defaultPolicy` (at most 100).

3.11 struct PolicyChange

Field	Type	Description
policy	ManagementPolicy	Policy to grant (POST).
expected	ManagementPolicy	Stored policy as last read; required to replace or revoke.

3.12 struct PolicyView

Field	Type	Description
entries	List	Editable local management policies.
readOnlyEntries	List	Provider-owned local policies.
collectedAt	DateTime	Collection time.

3.13 struct Removal

Field	Type	Description
removed	Object	Identifier of, or `true` for, the removed entry.
id	Identifier	Removed policy (policies only).

3.14 struct RulePriorityChange

Field	Type	Description
priority	Number	New priority, 1 to 2147483647.
expected	OrchestrationRule	Rule as last read (409 if changed, 428 if missing).

3.15 struct SoSChange

Field	Type	Description
configuration	Object	SoS model: `plantDescription` (name), `active`, `include`, `systems` with ports, `connections`.
expected	Object	Stored composition as last read (replace and remove).
activationConfirmed	Boolean	Must be `true` to save an active composition (428 otherwise).

3.16 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.

4 References

[1] J. Delsing, “LocalCloudManager – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-LocalCloudManager-java-spring/documentation/v5.2.2/LocalCloudManager-v5.2.2-SysD.pdf.

[2] ——, “localCloudManagement HTTP/TLS/JSON – Interface Design Description (LocalCloudManager),” Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudManager-v5.2.2-localCloudManagement-HTTP-IDD.pdf.

[3] ——, “localCloudConfiguration – Service Description (LocalCloudConfigurator),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudConfigurator-v5.2.2-localCloudConfiguration-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): console operations outside the registered list (features, authorization policies, rule priority, SoS, local-cloud configuration, chiplet floorplans) added; information model extended.	Claude Opus 5.5 (draft)
3	2026-11-09	5.2.2	Java-Spring implementation: registered as `generic_https/CERT_AUTH` on the Arrowhead connector with identity, whitelist and grant checks; session, login and logout console-only; local-cloud configuration through the orchestrated `localCloudConfiguration` service.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	